Elite Summer School

Digital Twins and simulation for robotic systems

Christian Schlette
Aug 2025

10 Aug 2025

Agenda

10:15	**Lecture 1**	Digital Twins	1.1 Digital Twins - definition(s)
			1.2 DTs for decision makers
11:30	*Lunch break*		
12:30			1.3 Examples of DTs in automation
			1.4 Deep dive: DTs for LEGO's production
			1.5 DTs for engineers
14:00	**Lecture 2**	Simulation	2.1 Types of simulation
			2.2 Methods in simulation
			2.3 Performance of automated production
15:00	*Coffee break*		
15:30	**Lecture 3**	Large Structure Production	
16:30	*End of the day*		

Digital Twins

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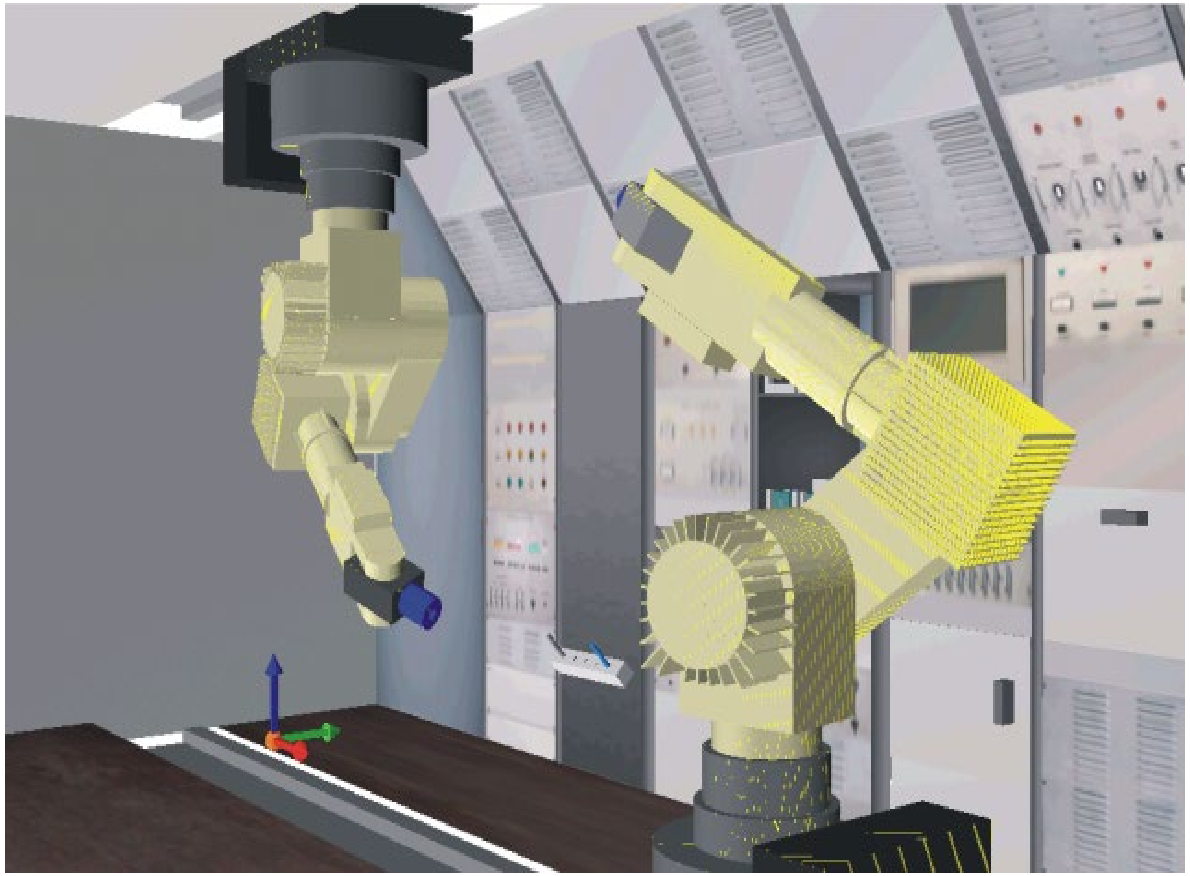
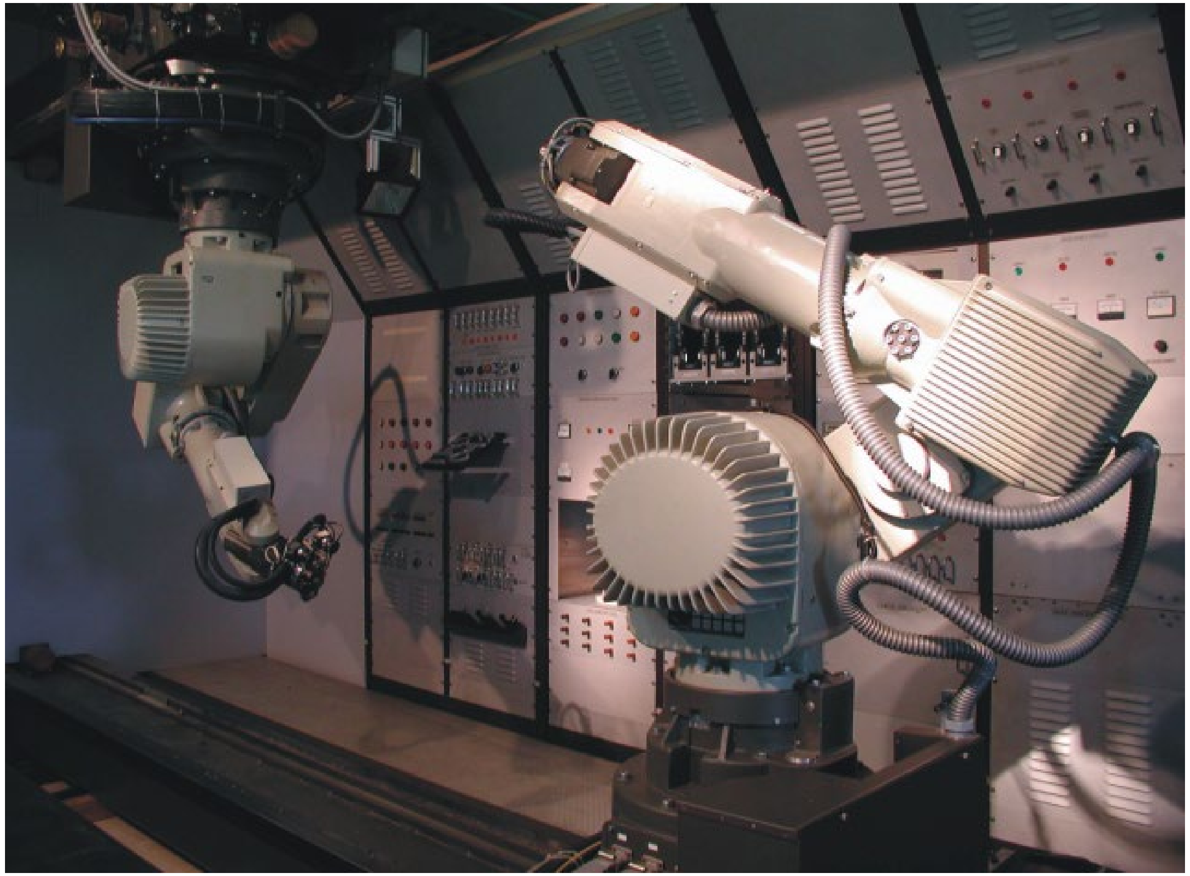
Aug 2025

Robot simulation and robot control

sigh … my first “digital twin”





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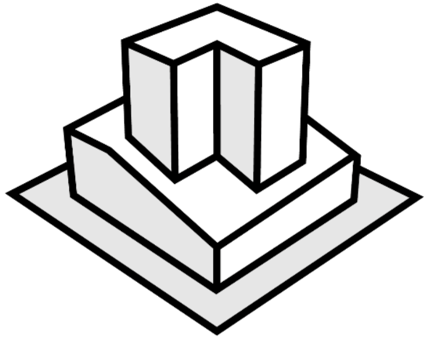
Overview

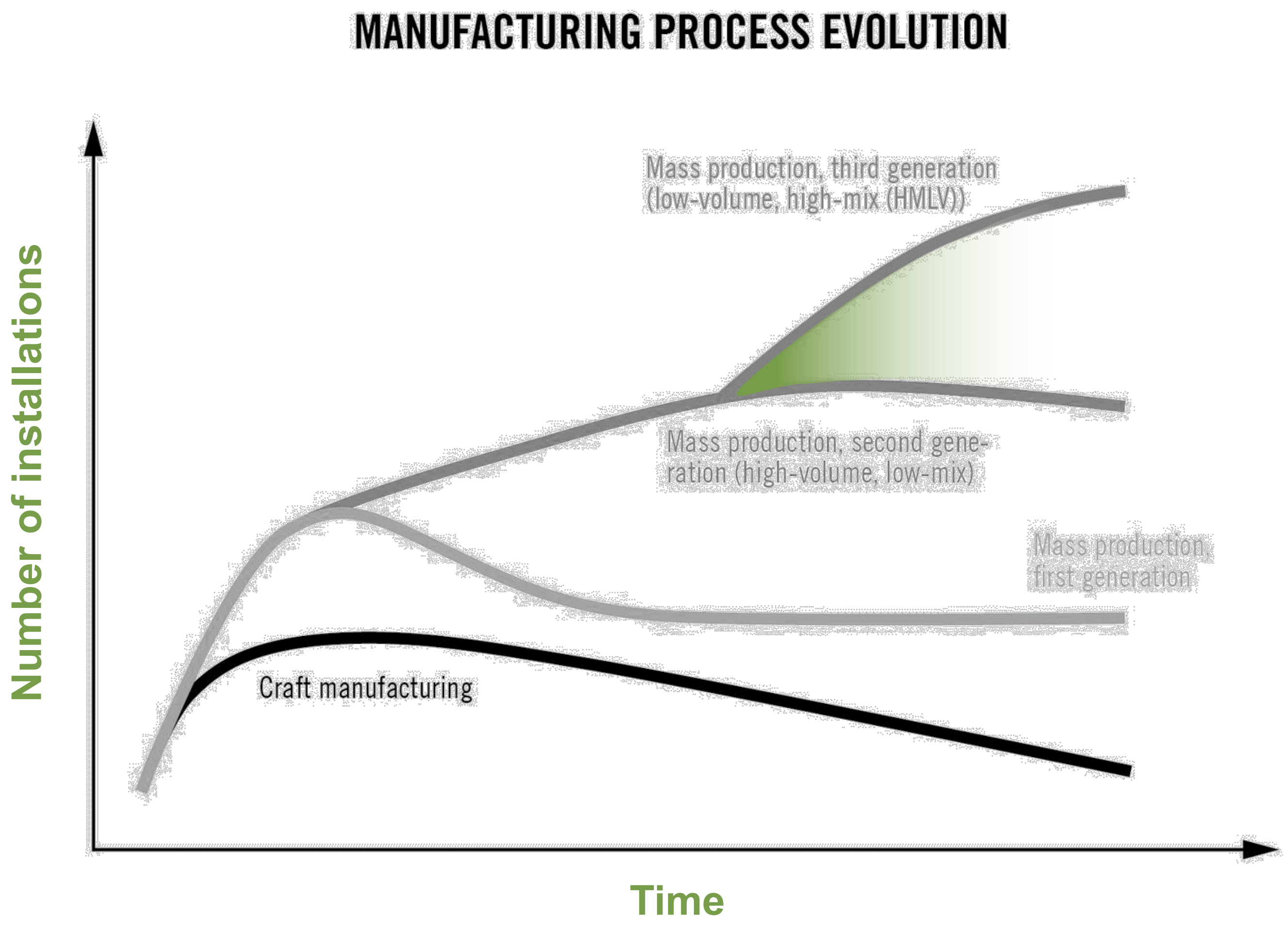
Lecture 1 Digital Twins

1.1 Digital Twins – definition(s)
1.2 DTs for decision makers
1.3 Examples of DTs in automation
1.4 Deep dive: DTs for LEGO’s production
1.5 DTs for engineers



Digital Twins – definition(s)

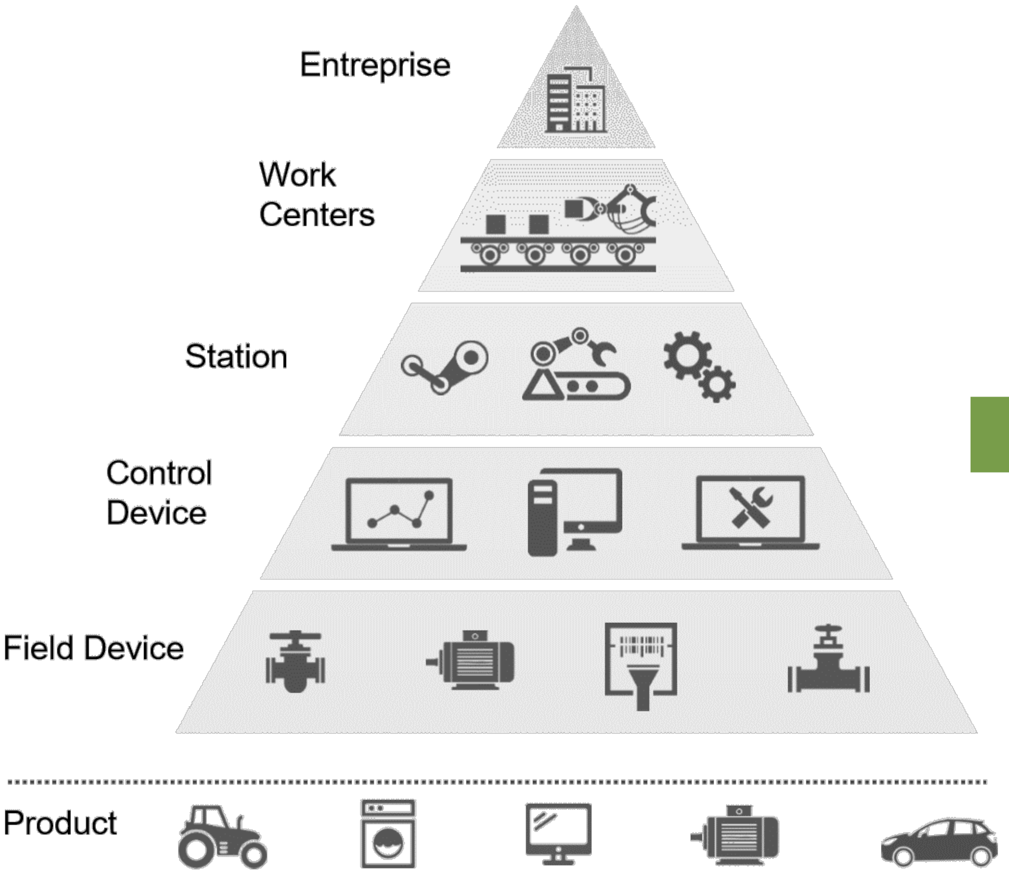
Why DTs?

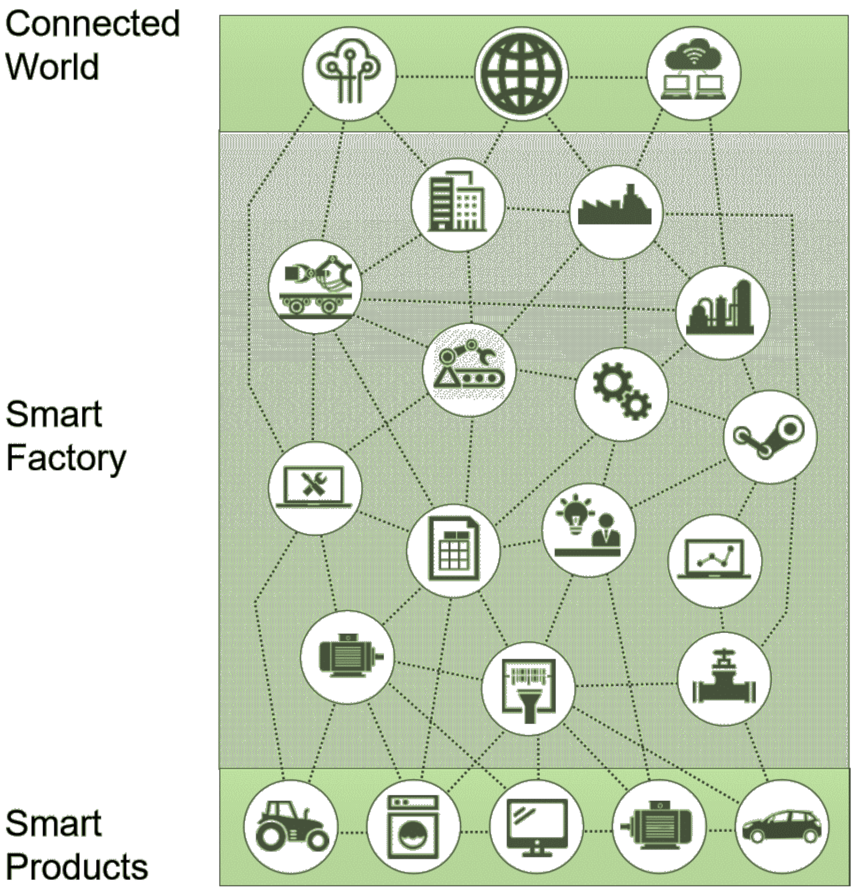


Source: SECO Tools 2019

Breaking existing hierarchies

Industry 4.0 redefines horizontal and vertical integration schemes





Source: Plattform Industrie 4.0 2016

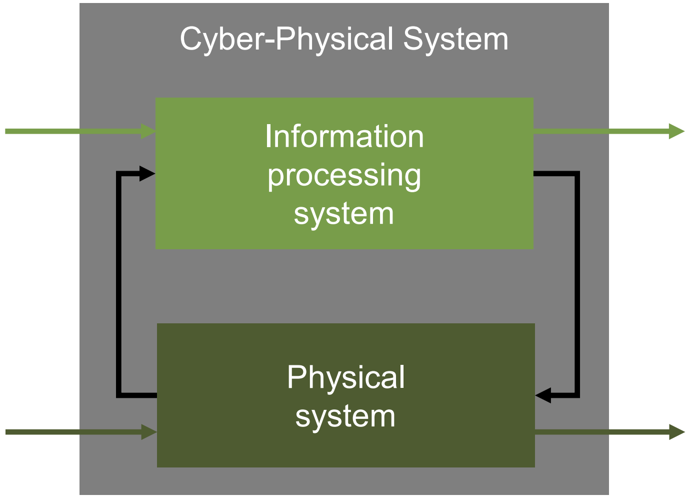
Cyber-Physical Systems (CPS)

Cornerstone of Industry 4.0

CPS connect the virtual world (cyberspace) to the real world.

  Powerful embedded systems (microcomputers)
+ Wireless networking on the Internet
+ Huge address-space with IPv6 (2012)
+ Decreasing costs for sensors and memory

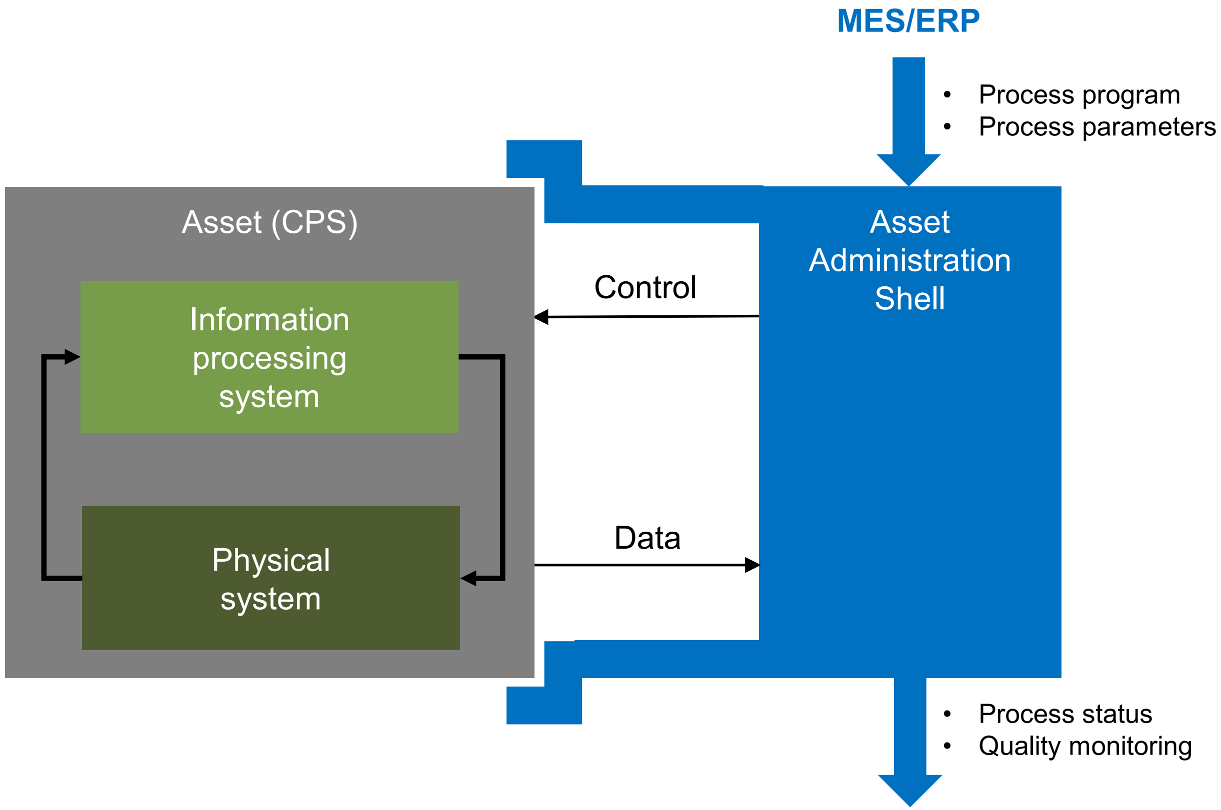
= Internet of Things (IoT) and Services



Khaitan, S. K. ; McCalley, J. D.: Design techniques and applications of cyber physical systems – a survey. In: IEEE Systems Journal, 2015

Data-oriented definition of DTs

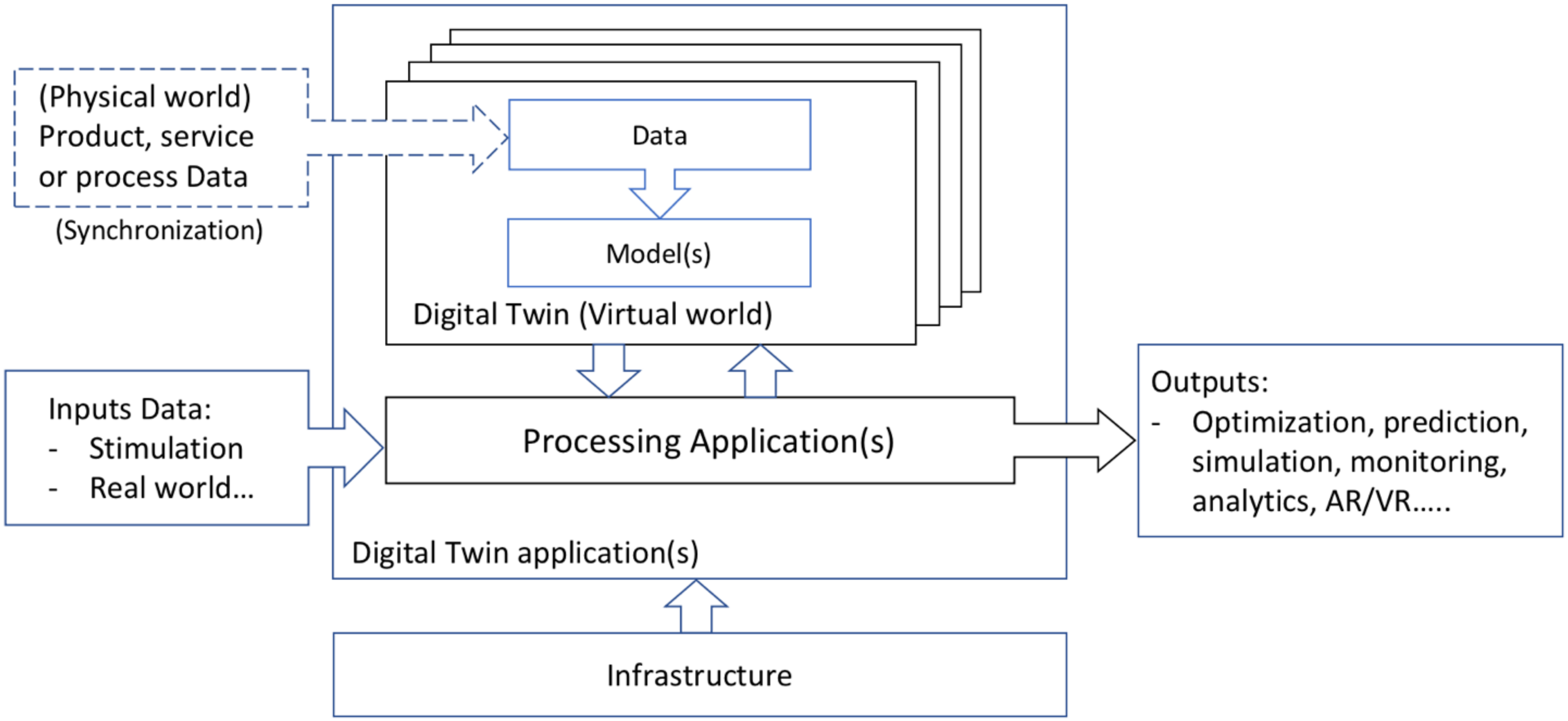
Asset Administration Shell



Plattform Industrie 4.0: Asset Administration Shell Specifications, 2020
Grieves, M., Virtually intelligent product systems: Digital and Physical Twins. In: in Complex Systems Engineering: Theory and Practice 2019

ISO definition of DTs

Functional view and model processing for outputs



Norm ISO/IEC NP 6518, 2020

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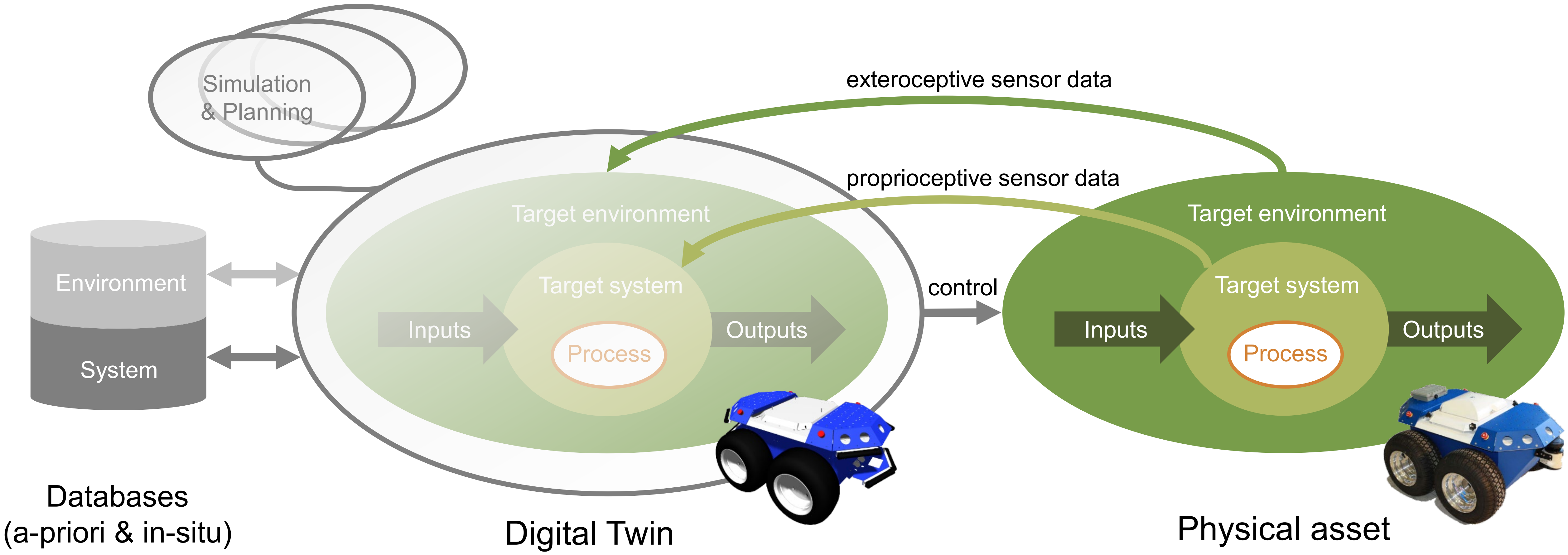
Digital Twins

System-oriented view in robotics



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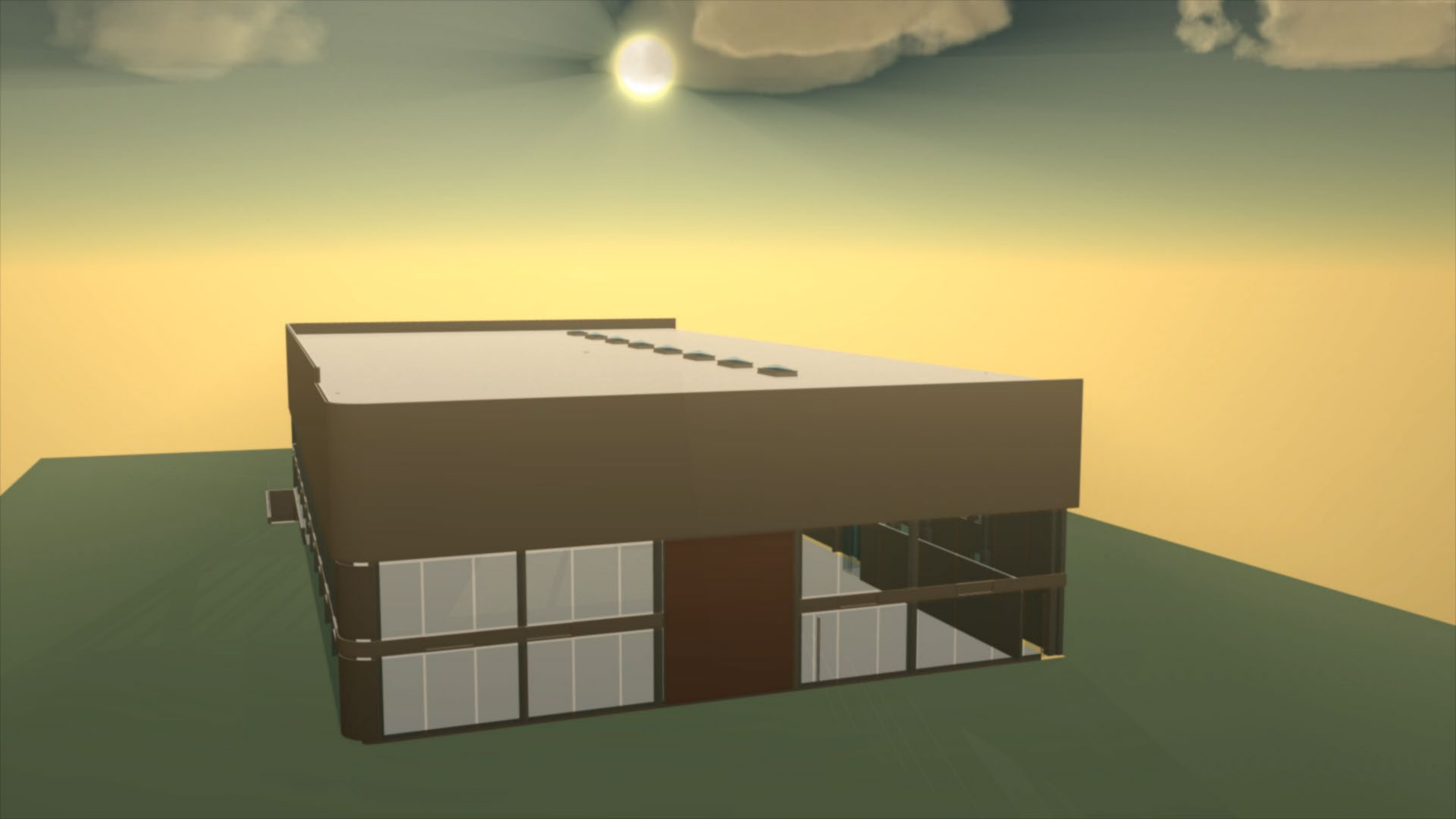
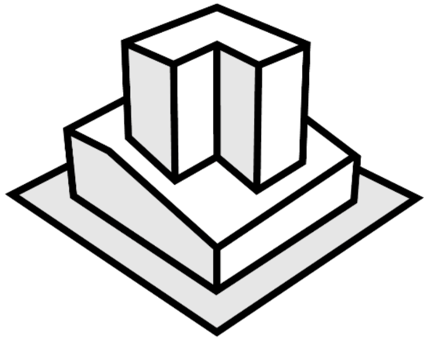


Digital Twins
Exercise (5 min.)

Think of another automation system.

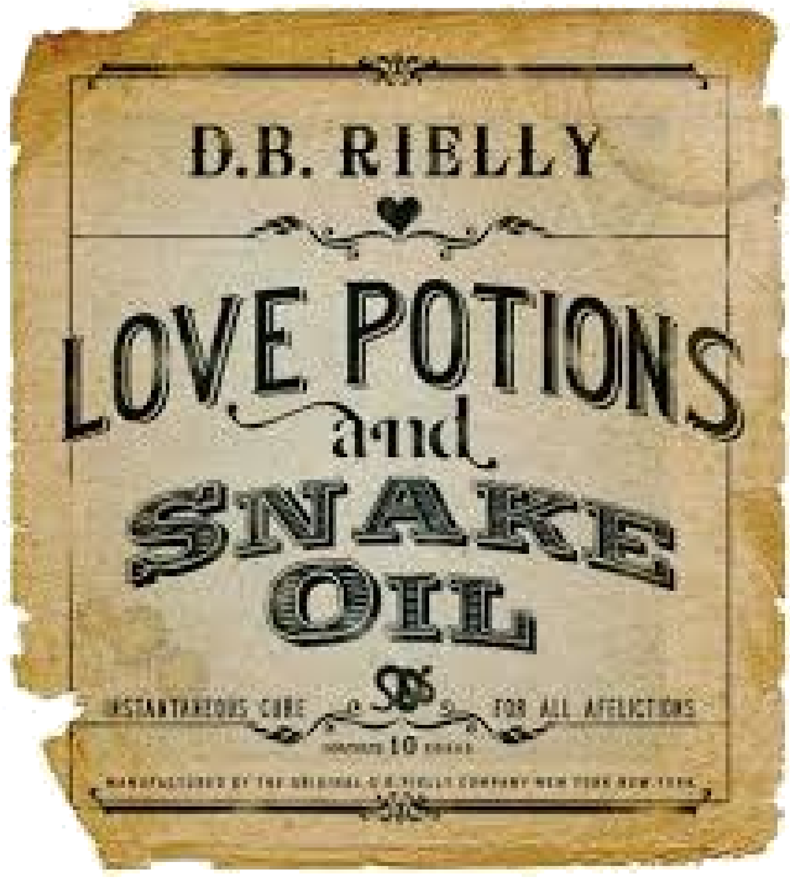
1. What is the target environment?
2. What could be a process one might be interested in for this system?
3. What could be inputs/outputs in the exchange with the target environment?
4. What could be examples of proprioceptive/exteroceptive sensors?
5. What could be a simulation one might be interested in?



DTs for decision makers

Digital Twins

The snake oil of engineering… (now replaced by AI ☺)



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What is (not) a Digital Twin?

Discussion (5 min.)

Think of our DT definition.

1. Is a 3D model a DT? Yes/no – and why?

2. Is a (3D) simulation a DT? Yes/no – and why?

3. Psst… has some tried to sell you a “DT” in the past?

DTs for decision makers

Mini-guide / Essential questions

- DTs need physical counterparts. That can be your products, your production systems or your production environment.
- DTs that only read data from physical counterparts are called “Digital Shadows”. There are not “true” DTs, but… tolerated.
- Pure (3D) models or (3D) simulations are not DTs. Don’t invest in solutions making such claims, unless you know what you want.
- Know what you want:
  - What is your physical asset?
  - Why do you need a DT? (Or do you actually need a simulation?)
  - Which data do you want read from the physical asset and how?
  - Which control do you want take over the physical asset and how?
- And if you decide for a solution – do you have all information and know-how to model/program/maintain/extend it in the future?

DTs for decision makers

Mini-guide / Added value

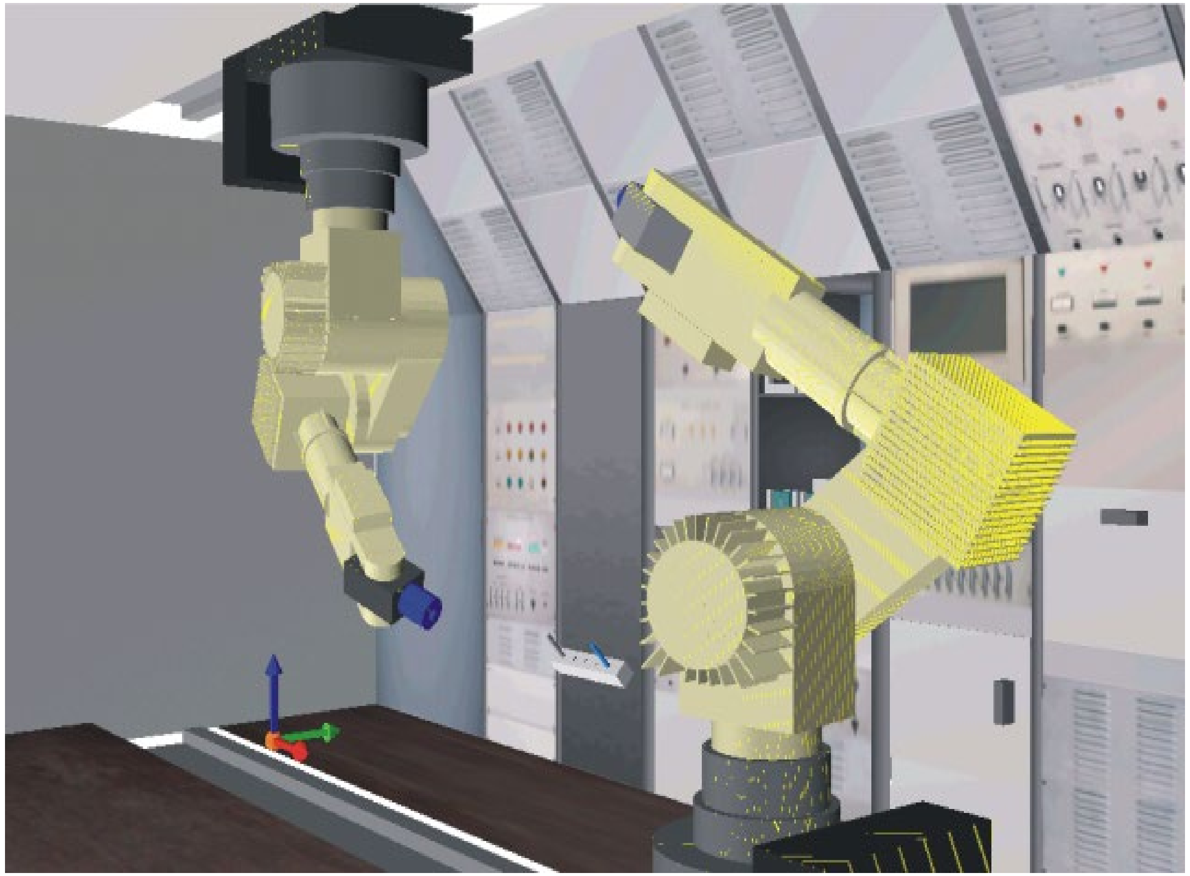
- DTs add value by providing engineers a solution along the engineering process of an asset:
  - System design
  - System verification
  - Component programming
  - System integration
  - System validation

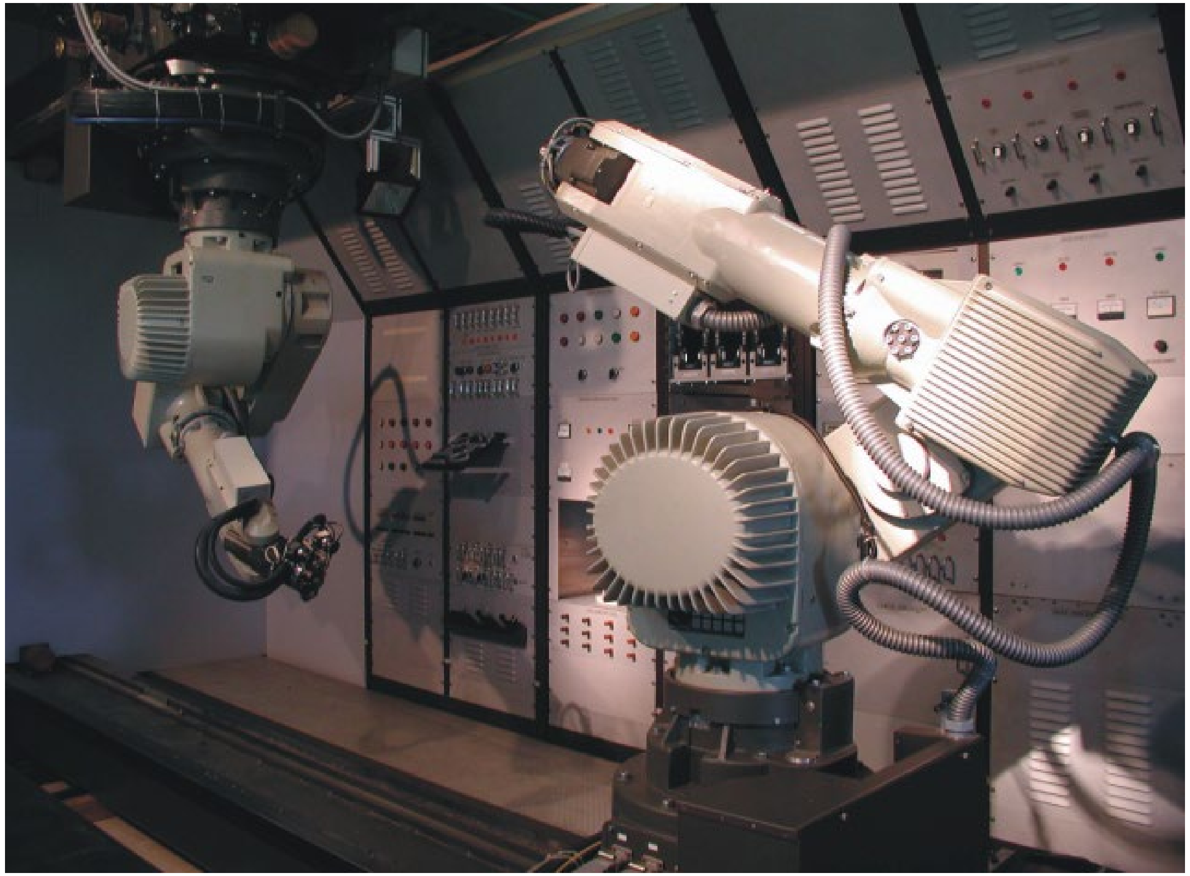
- Beyond engineering, DTs add value by reusing the models, simulations, data, etc. throughout the asset lifecycle:
  - Operation and optimization
  - Testing and maintenance
  - Extension and replacement
  - Training

Examples of DTs in automation

Robot simulation and robot control

Test setup for robotic operations on a space station

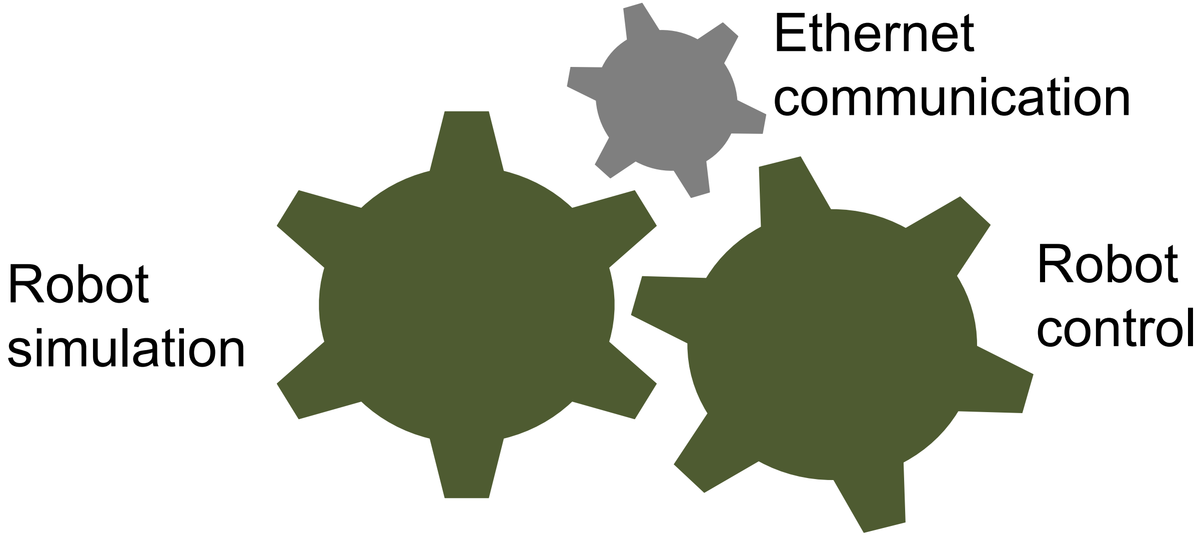





CIROS

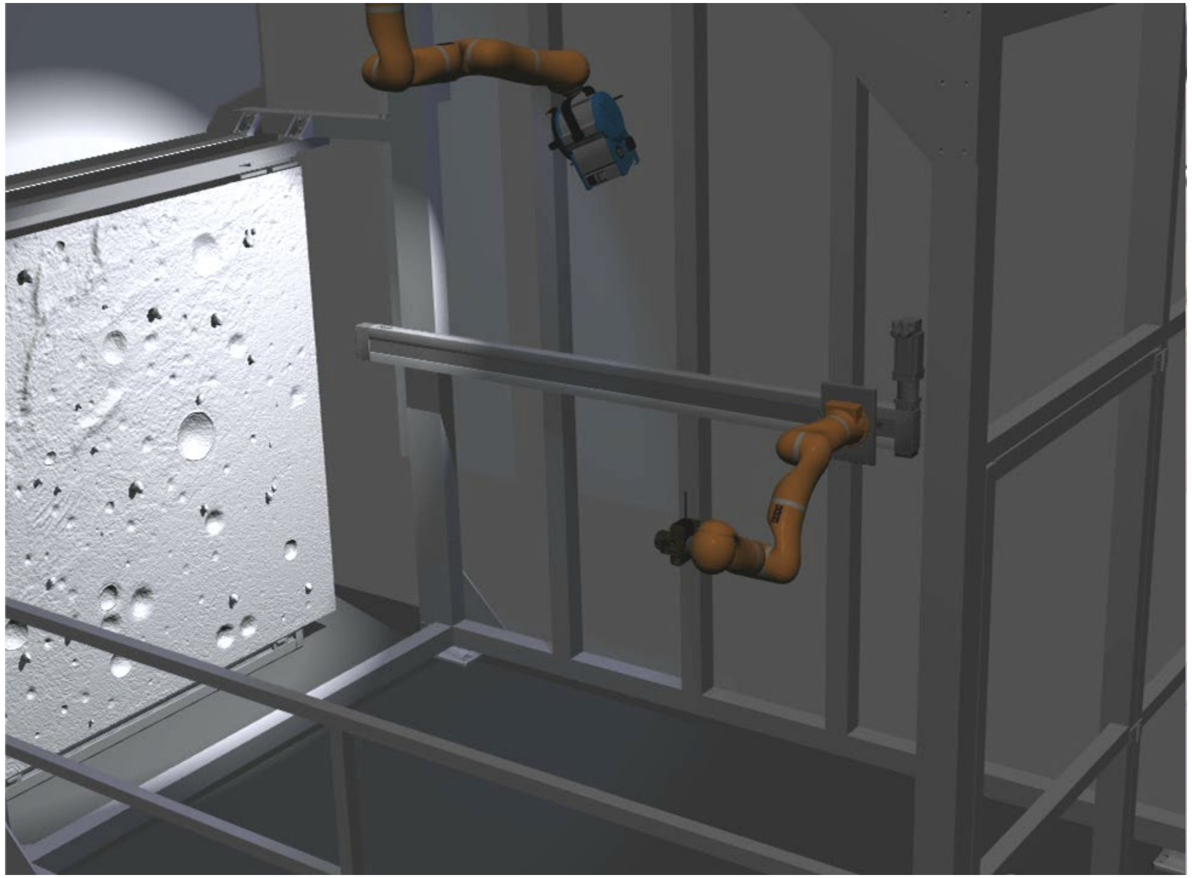
System components

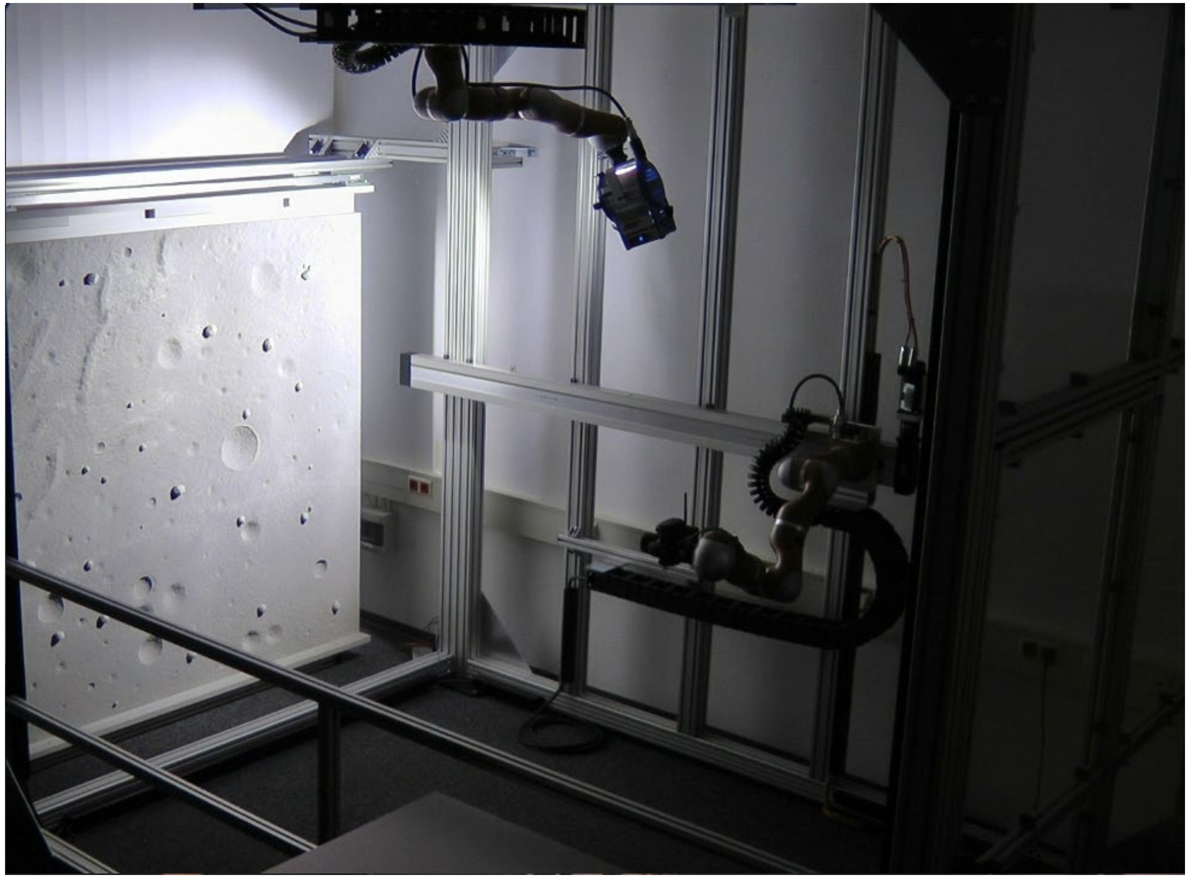
Robot simulation, monitoring and control



Simulation-based system development

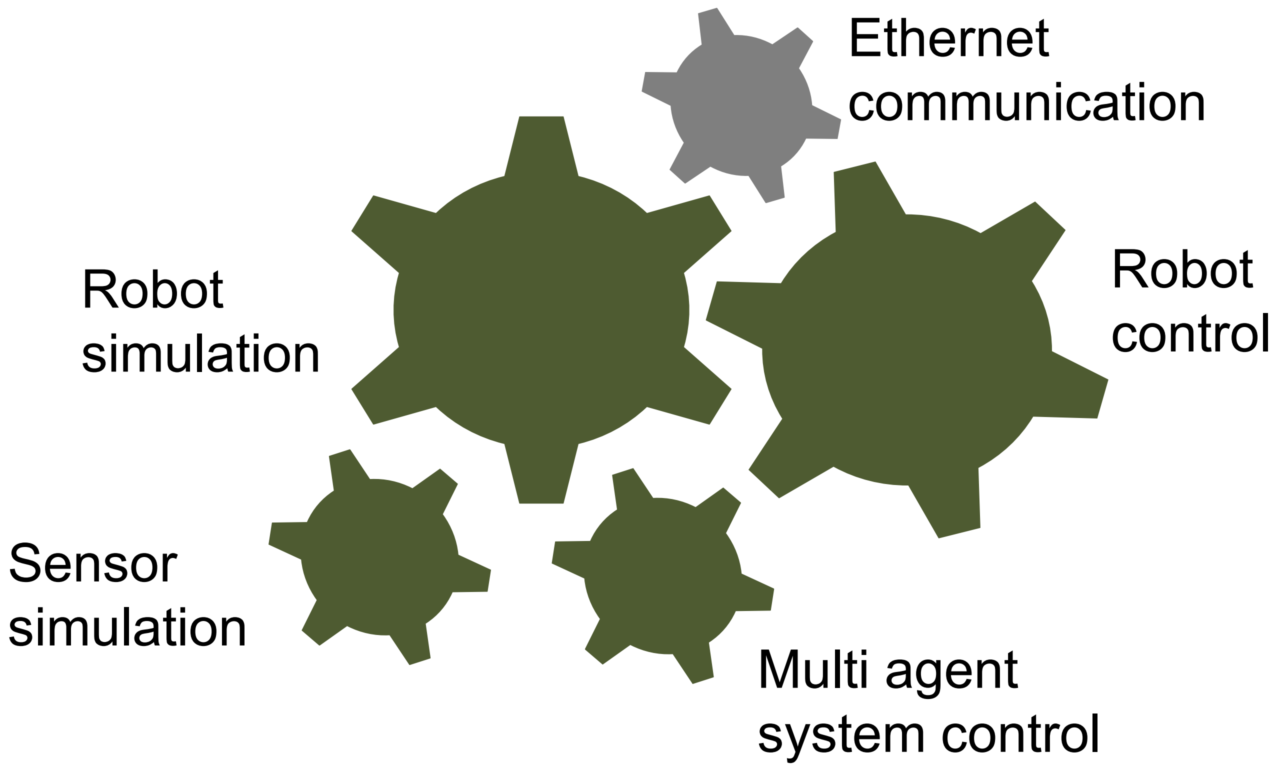
Mockup for optimization of approach trajectories for DLR





System components

…plus sensor simulation and MAS control

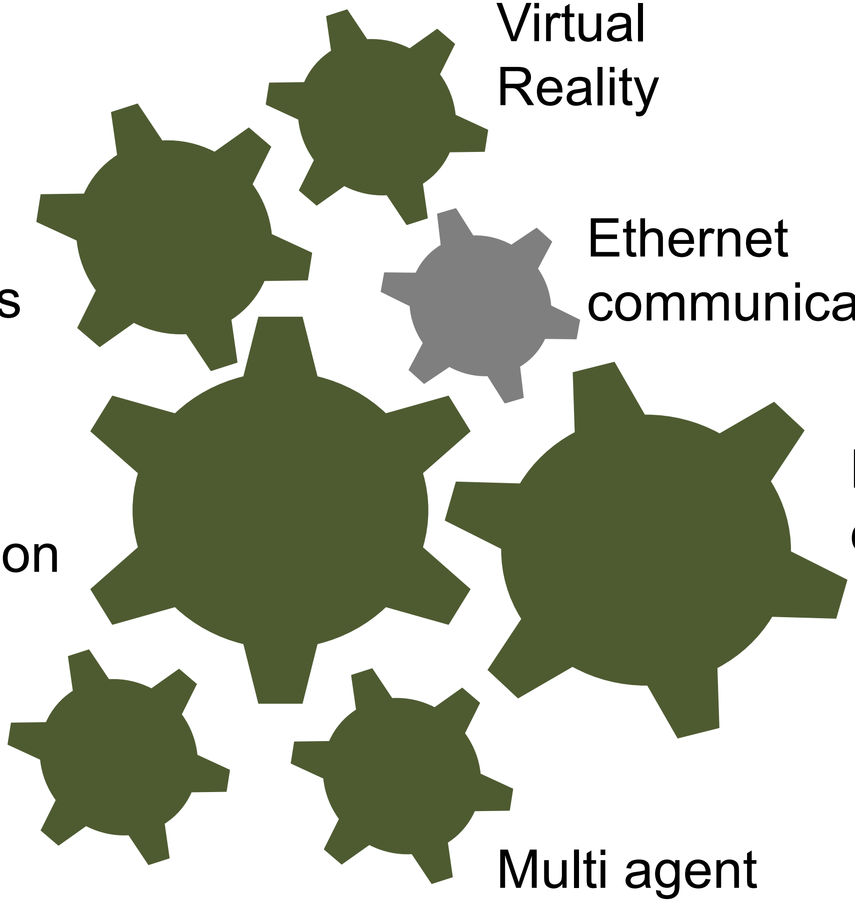


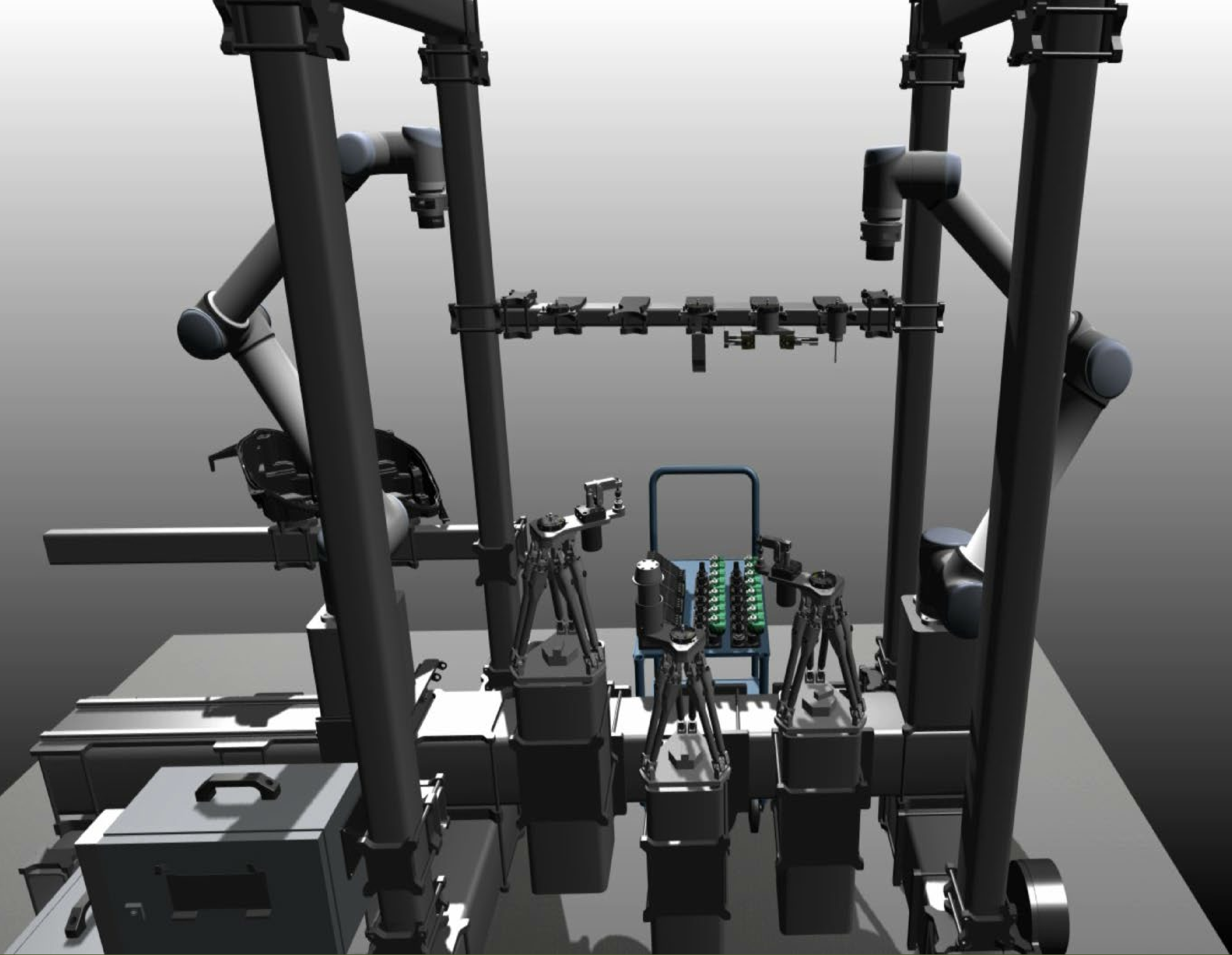
Simulation-based system development

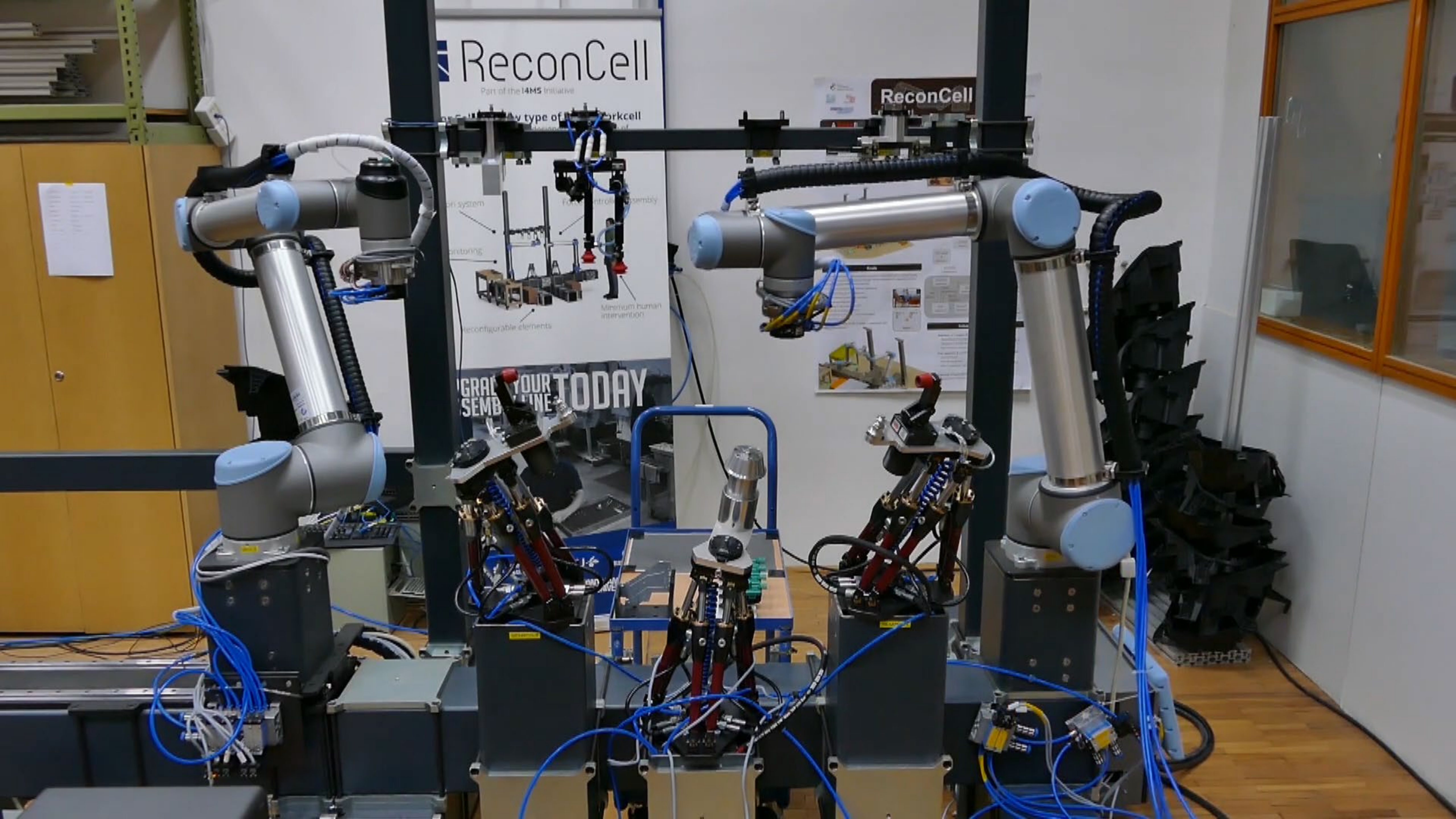
R&D and operator training for the state of North-Rhine Westfalia (NRW)

System components

DB technologies and VR to pull in, manage and visualize large environments

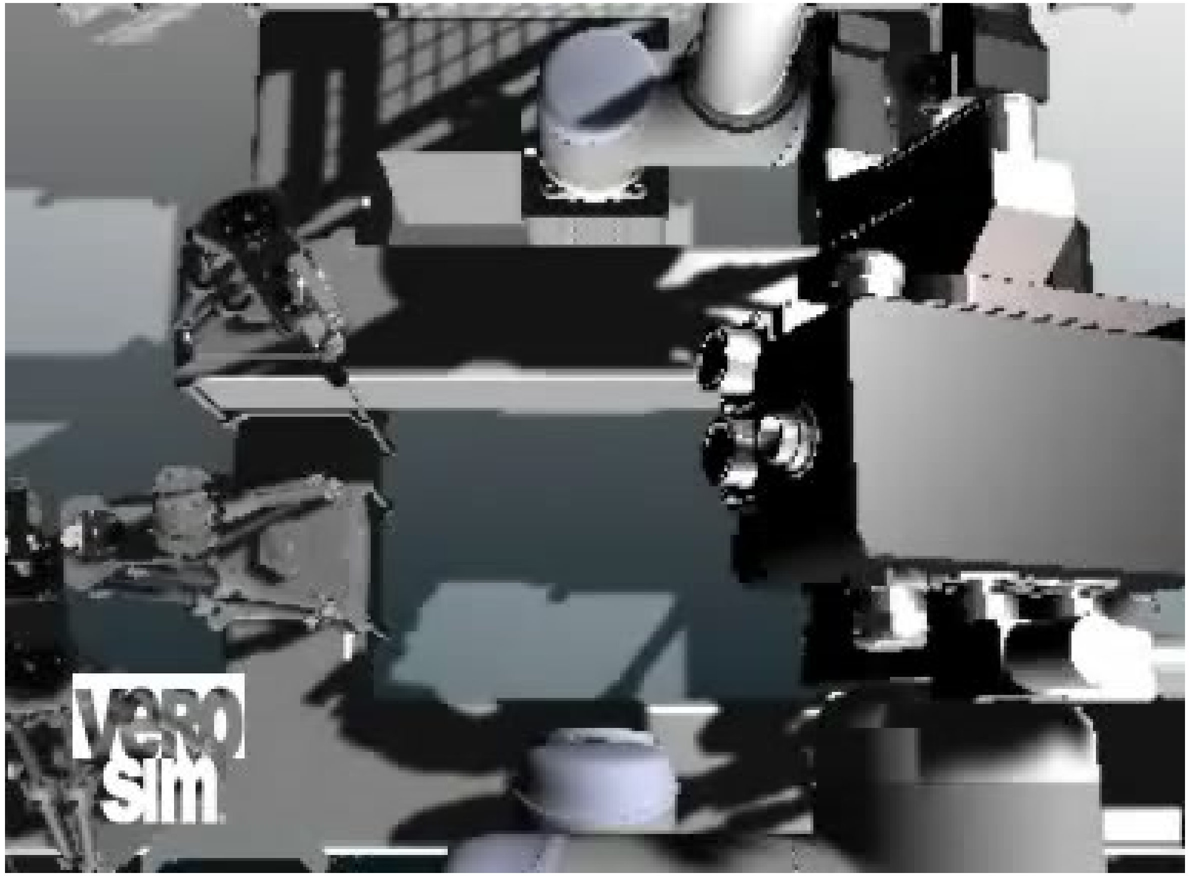


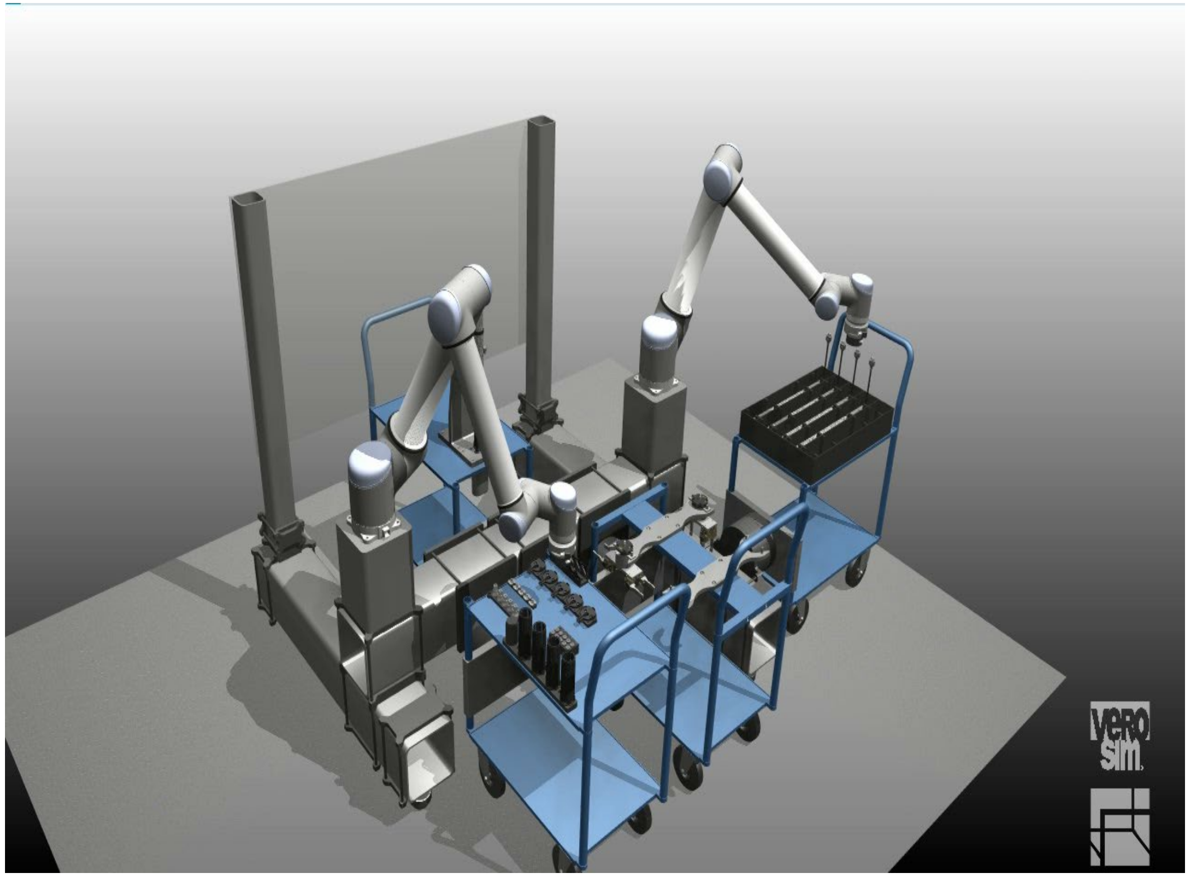



ReconCell
Part of the I4MS Initiative
Minimum human intervention
TODAY
ReconCell

Optimization of equipment poses

e.g. camera and robot positioning in ReconCell





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Deep dive: DTs for LEGO's production

14 – 16 MARCH

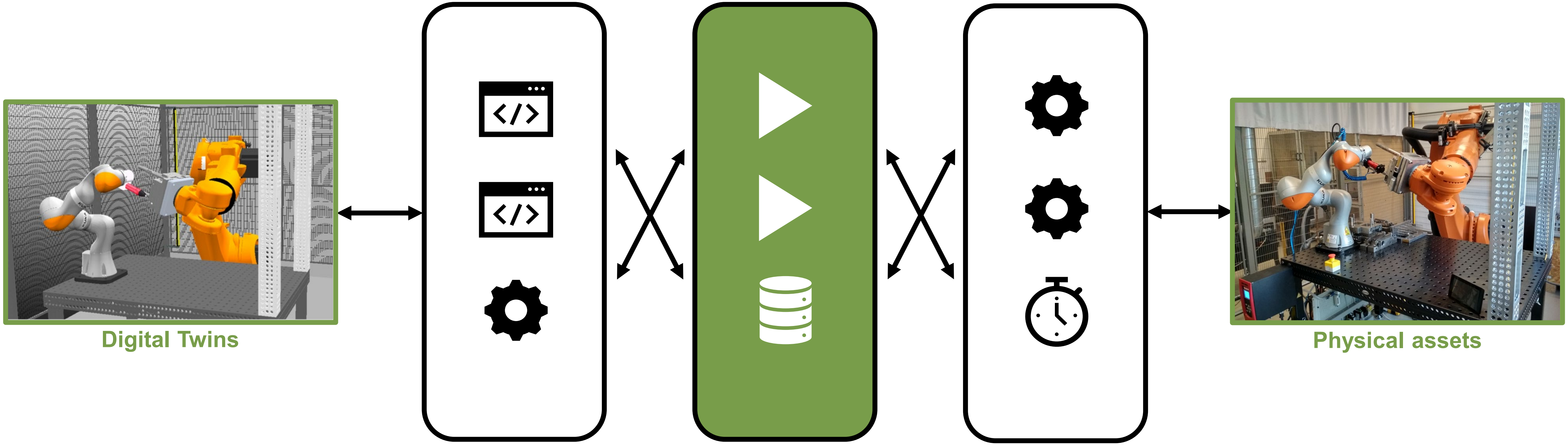
Odense • Denmark

ERF2023

EUROPEAN ROBOTICS FORUM

Service-oriented architecture

Linking Digital Twins to physical assets



Planning layer
user interfaces,
simulation and
high-level control

Abstraction layer
service registry
and databases

Real-time layer
controllers and
low-level control

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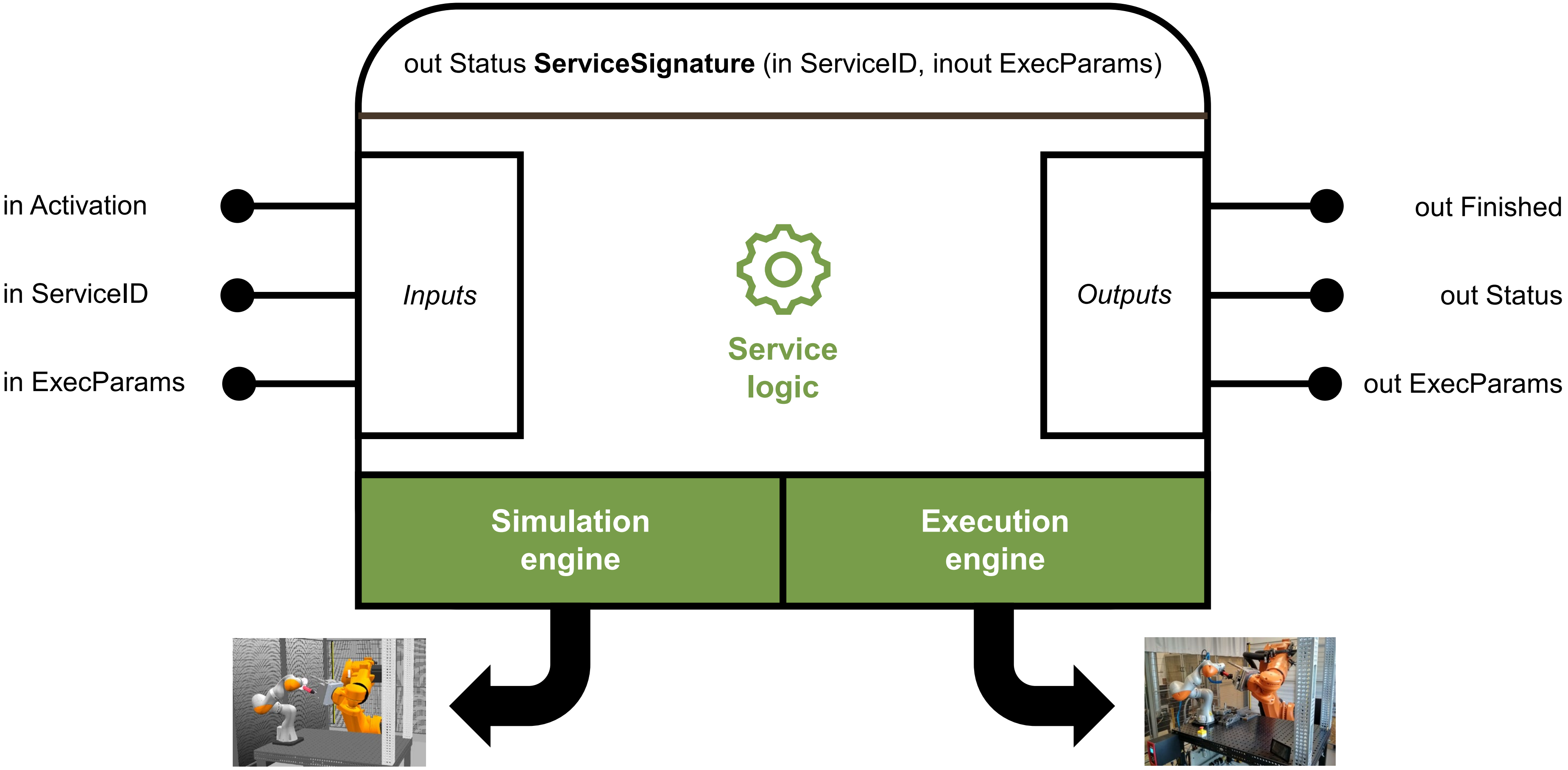
Visual programming

ServiceBlocks to encapsulate expert knowledge



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Wow, what a DT!

Exercise (5 min.)

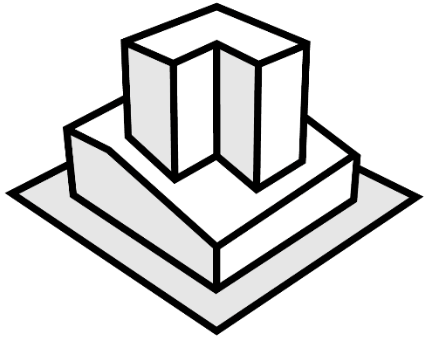
Think of the LEGO case.

1. What are the physical assets?

2. Which functions does the DT have?

3. Which data do you think is exchanged between assets and DT?

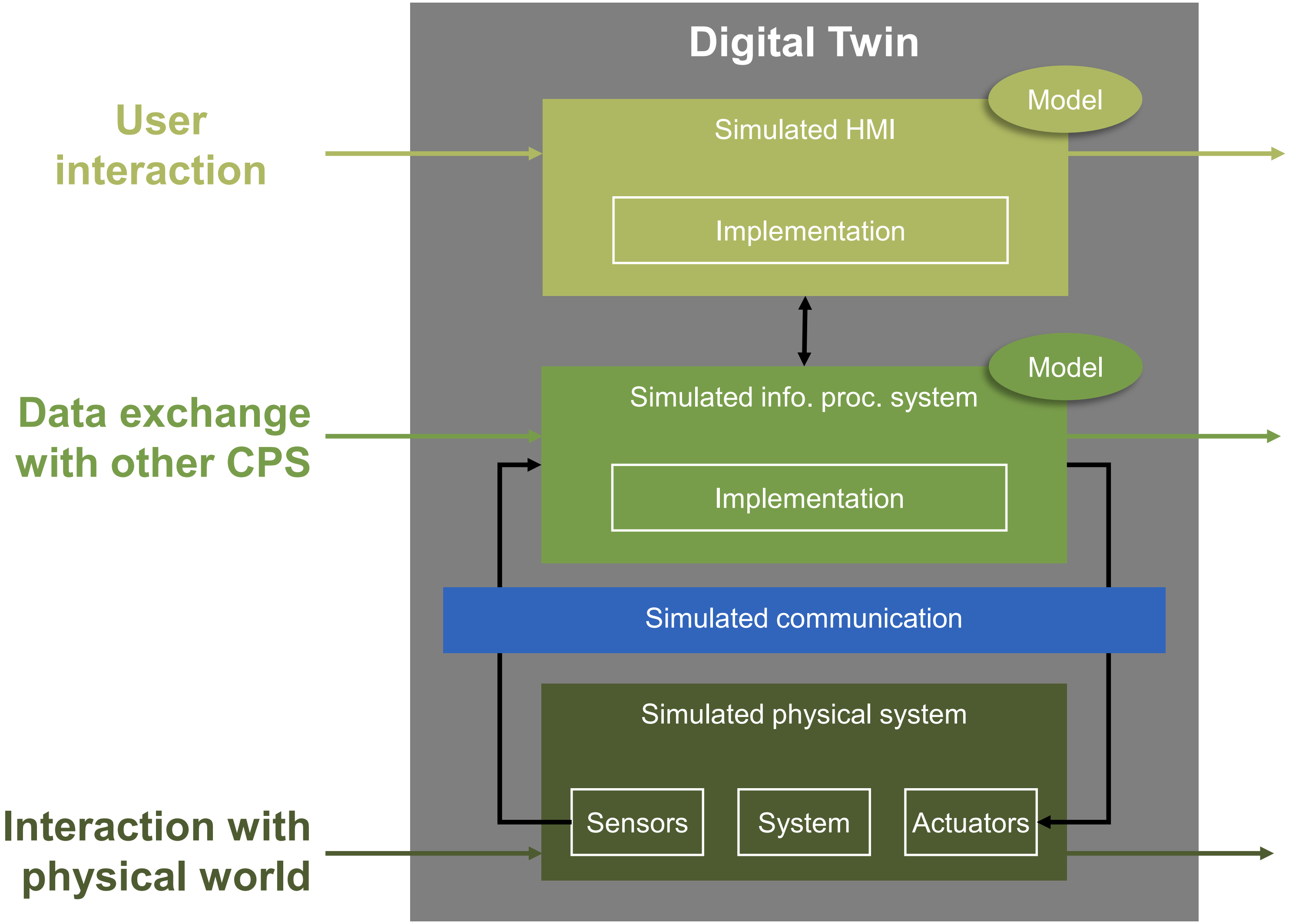
4. The operator has a DT too – what data is exchanged between operator and DT?



DTs for engineers

Systematic description of DTs

Digital Twin



Schluse, M. ; Priggemeyer, M. ; Atorf, L. ; Rossmann, J.: Experimentable Digital Twins – Streamlining simulation-based systems engineering for Industry 4.0. In: IEEE Transaction on Industrial Informatics, 2018.

DTs for engineers
Mini-guide / Planning

- In the planning phase:
  - List all assets, list all desired DT functionalities.

  - List the necessary versus all available data streams from asset to DT to monitor the asset.
  - How can the data streams be interfaced and transmitted?

  - List the necessary versus all available control functions the DT calls to control the asset.
  - How can the control functions be interfaced and transmitted?

  - Collect all digital models of your assets – spec sheets, CAD data… just all.

- Identify a commercial DT solution to support the necessary interfaces and import options.
- Be ready to compromise and couple various solutions.

- Consider using System Modeling Language (SysML) to plan your DT.

DTs for engineers

Mini-guide / Commissioning & operation

- Be clear – do you buy a readily commissioned solution or do you commission the DT in-house?

- If provided key-ready:
  - Invest in solid service, prepare for reoccurring services, avoid vendor lock-in.

- If commissioned and operated in-house:
  - Invest in software-oriented staff to model/program in the selected solution.
  - Invest in hardware-oriented staff to connect assets to the selected solution.

  - Keywords: PLC, OPCUA, MQTT, Profinet, Azure, C/C++, Python, ROS.
  - PS: Robotic students may have touched many of them in their education ☺

  - Watch out for new interface options for your assets, which may simplify your solution.

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Simulation

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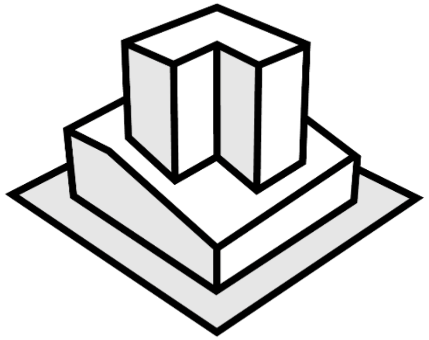
Christian Schlette
Aug 2025

Overview

Lecture 2 Simulation

2.1 Types of simulation
2.2 Methods in simulation

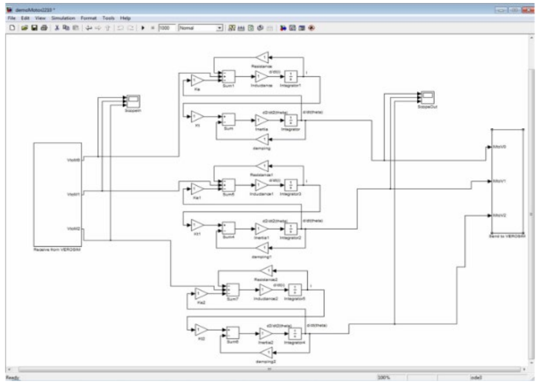


Types of simulation

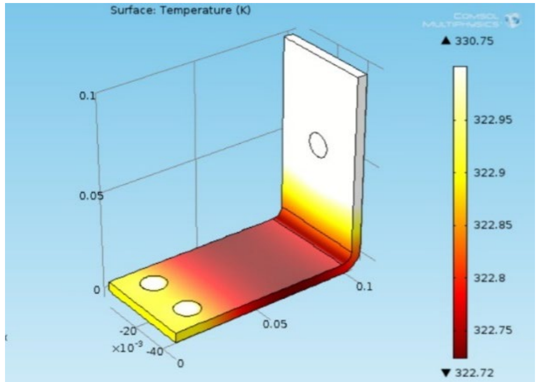
Types of simulation

Simulation in multiple domains

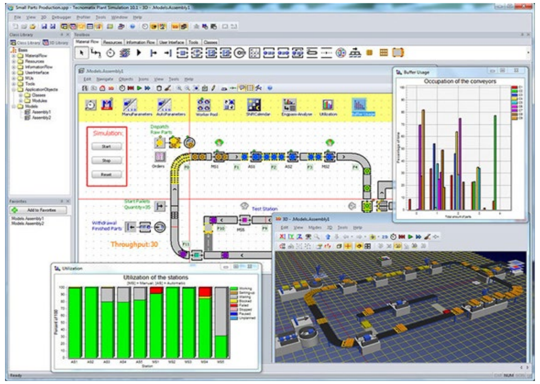
Block-oriented simulation
(e.g. Mathworks Simulink)



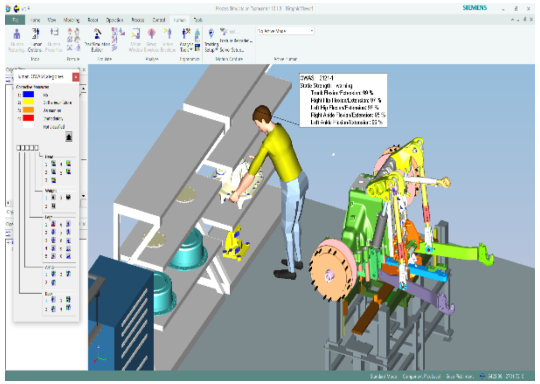
FEM analysis
(e.g. COMSOL FEM)



Discrete event simulation
(e.g. Siemens Plant Simulation)



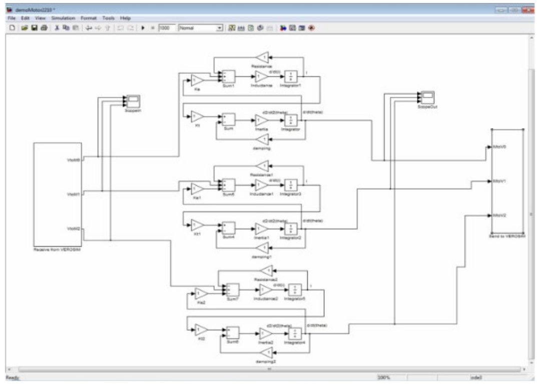
(Quasicontinuous) process simulation
(e.g. Siemens Process Simulation)



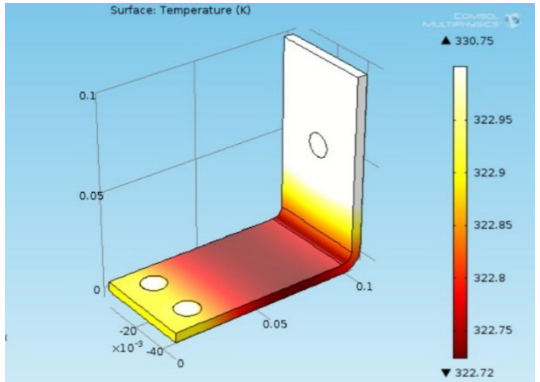
Interactive 3D simulation
(e.g. VEROSIM Harvester simulator)

Types of simulation
Simulation in multiple domains

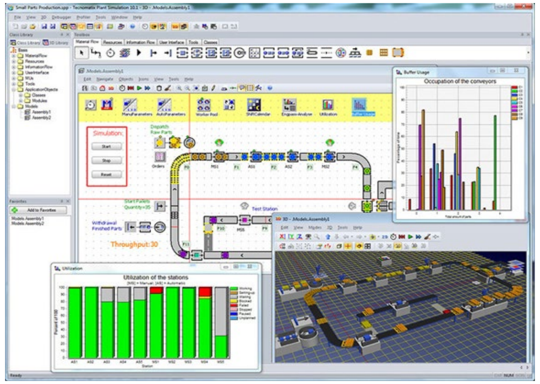
Block-oriented simulation
(e.g. Mathworks Simulink)



FEM analysis
(e.g. COMSOL FEM)



Discrete event simulation
(e.g. Siemens Plant Simulation)



(Quasicontinuous) process simulation
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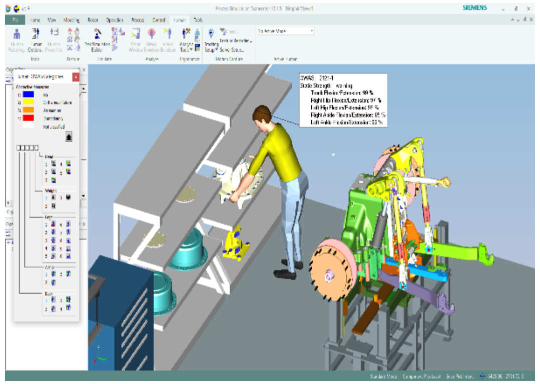
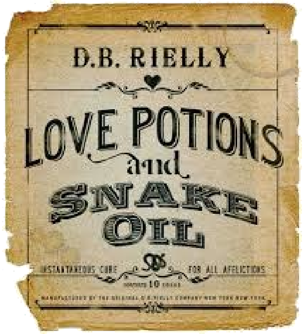


Interactive 3D simulation
(e.g. VEROSIM Harvester simulator)

Digital Twins…



Basic concepts

An ontology

Adaption of Schluse, M. ; Schlette, C. ; Waspe, R. ; Roßmann, J.: Advanced 3D Simulation Technology for eRobotics. In: Proceedings of the IEEE International Conference on Developments in eSystems Engineering (DeSE 2013), pp. 151-156. - ISBN 9781479952632, DOI 10.1109/DeSE.2013.35

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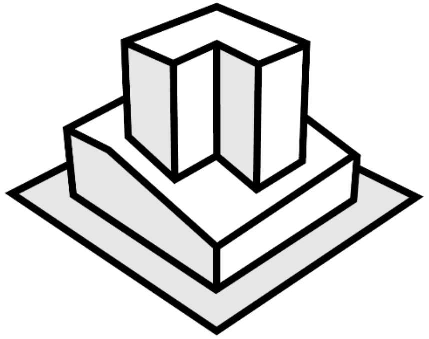
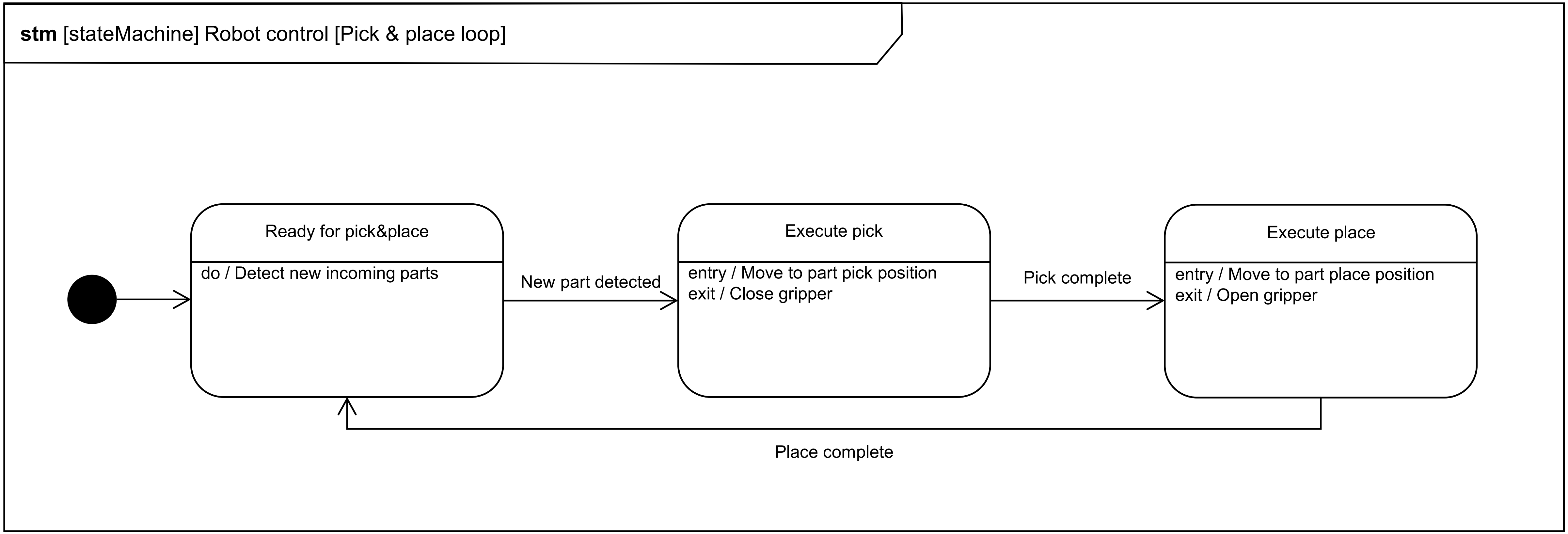


Methods in simulation

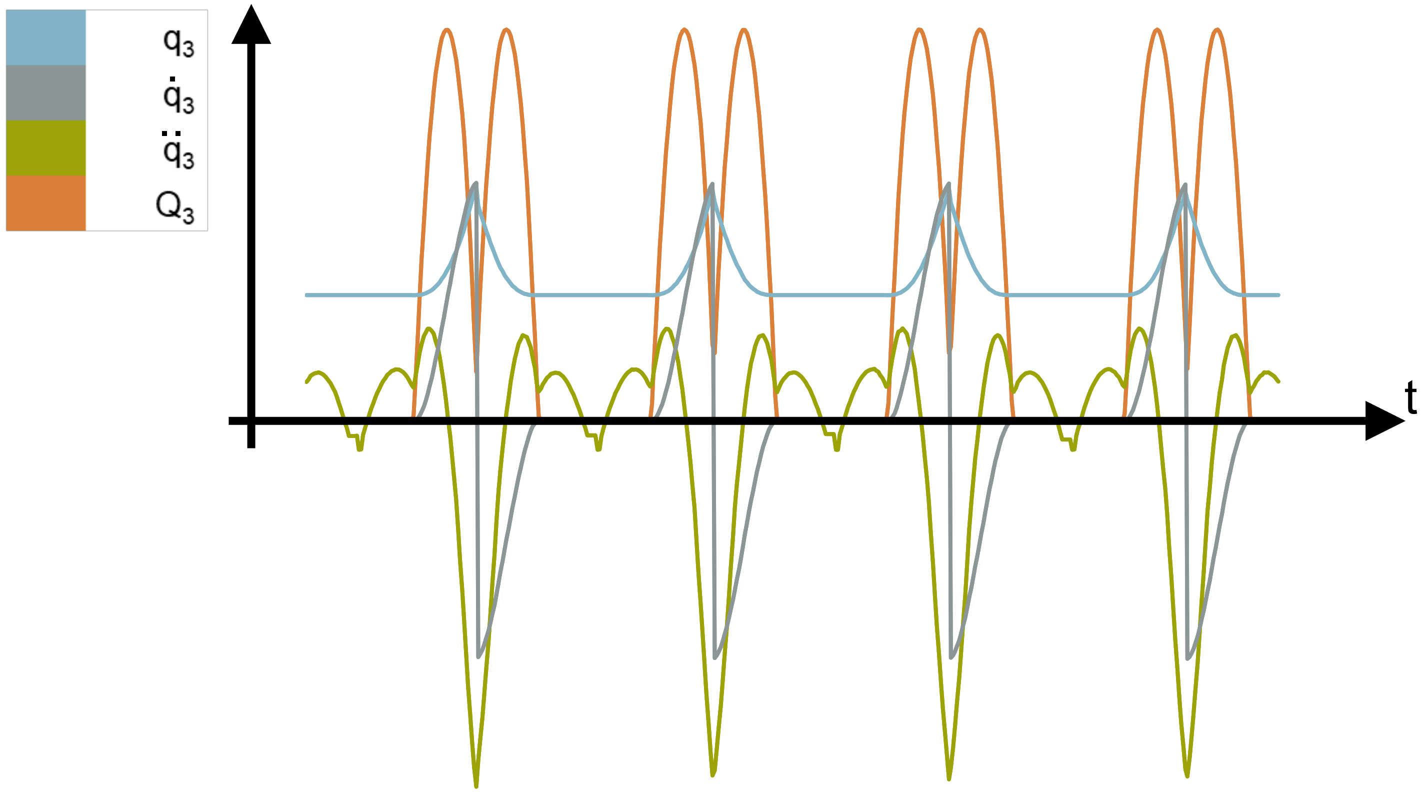
Simulation scheduling

Discrete event simulation



Simulation scheduling

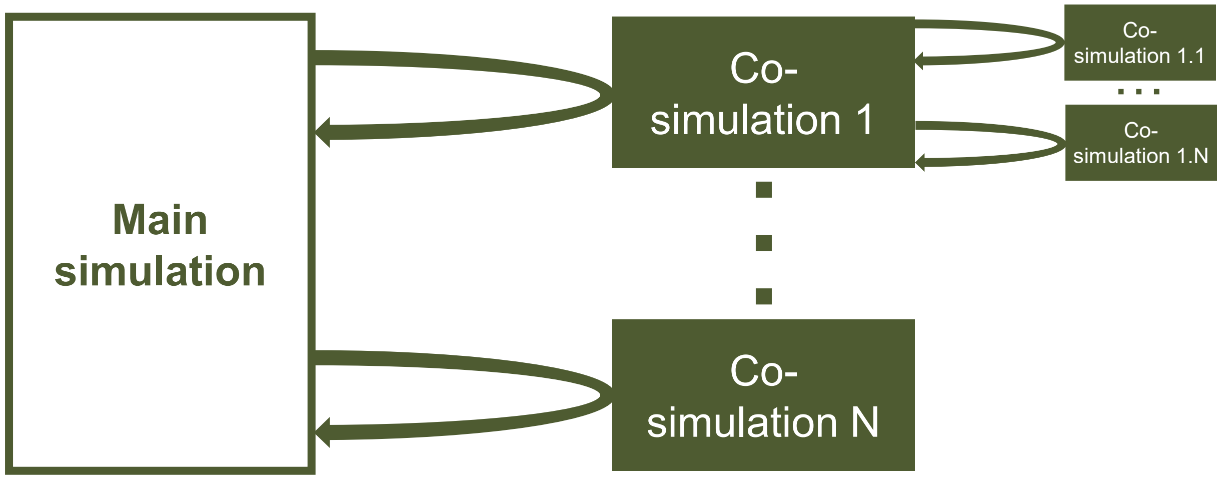
(Quasi-)continuous simulation



Schlette, C.: Anthropomorphe Multi-Agentensysteme : Simulation, Analyse und Steuerung. Wiesbaden : Springer Vieweg, 2013 - ISBN 9783658025182 (print), ISBN 9783658025199 (online), DOI 10.1007/978-3-658-02519-9

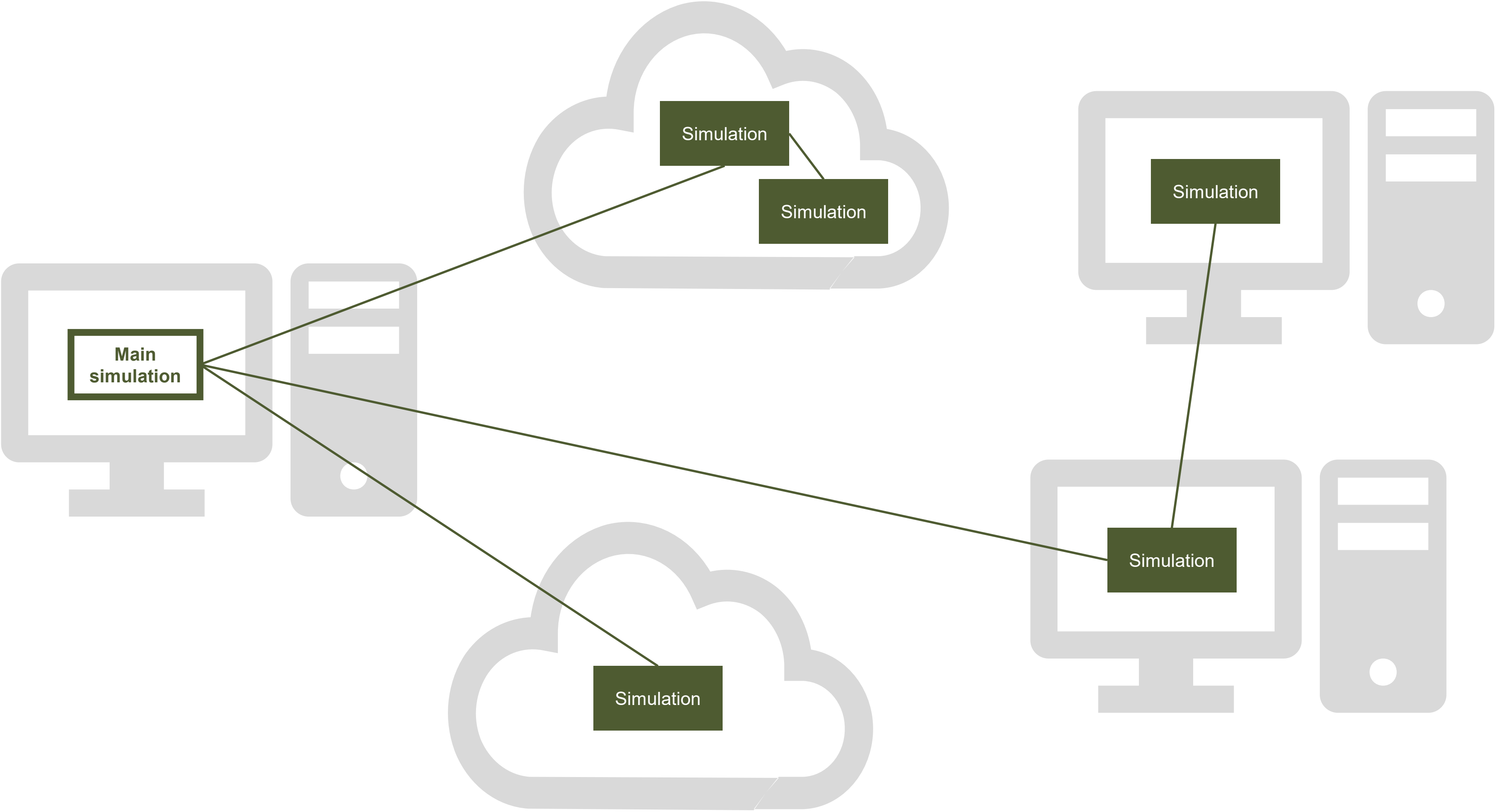
Simulation modes

Co-simulation



Simulation modes

Distributed simulation



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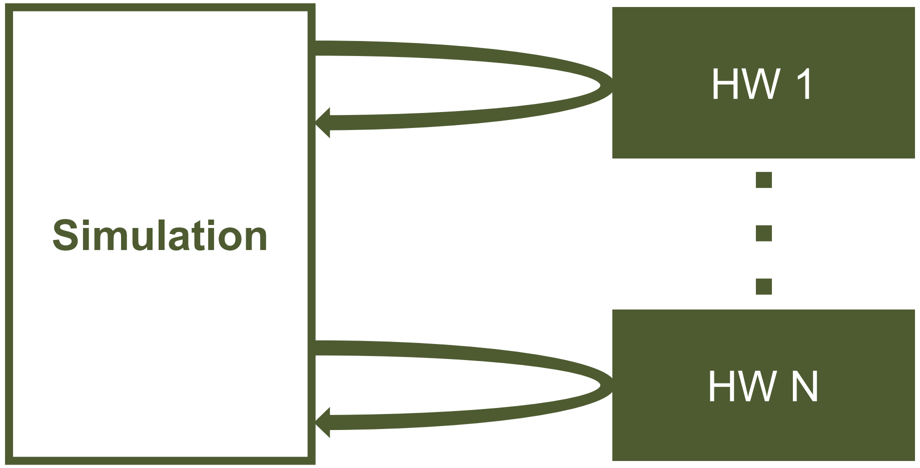
Simulation modes

Hardware in the loop



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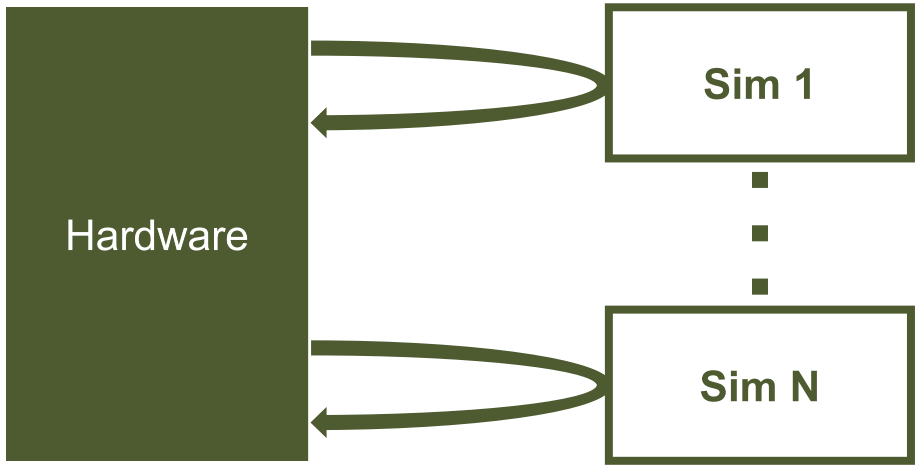
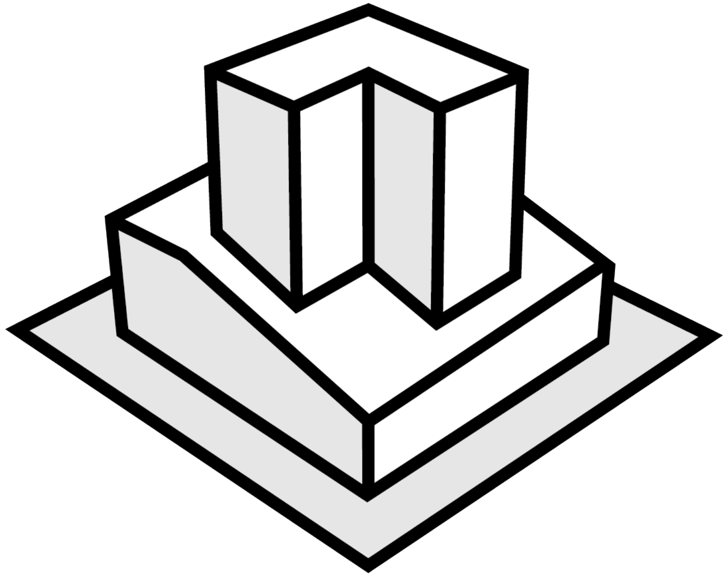
Simulation modes

Software in the loop



SDU Center for Large Structure Production (LSP)

Robotization and digitalization for the sectors maritime, construction & energy



Christian Schlette

Aug 2025

Christian Schlette
Professor

SDU Center for Large Structure Production (LSP)
The Mærsk Mc-Kinney Møller Institute (MMMI)
University of Southern Denmark (SDU)
https://lsp.sdu.dk

chsch@mmmi.sdu.dk
T +45 6550 7916
M +45 9350 7377